Standard recording 2 - Transcription

Speaker 1 00:00

They were all on the left side and then all the others were on the right side. But it's also only a two D layout. So it's always very diffi cult to have a layout that really nicely, cleanly separates families. What does it even mean to cleanly separate the icons? So I'm just wondering, um what's, ah so what's the exact?

Speaker 1 00:28

Of.

Speaker 1 00:31

The let's see, say the study, or what would be the result of your thesis? Where where do you see the main result of thesis? it's like t echnical work that you performed a lot so far. Ah amazing.

Speaker 1 00:45

Like so many icon datasets and and different measures and layouts and the dashboard and all that. But what do you see as a goo d outcome of your research? What would you like to show in the end?

Speaker 1 01:00

What is the result that you're aiming for?

Speaker 2 01:03

So for the results part, the, and the main aim is, ah we have the study, ah with the icons and we we see we can have some qualitati ve score in the qualitative skills, ah, um for the, ah icon's perception. So for example,

Speaker 2 01:22

maybe since we have seven feature families, maybe we can have some kind of ah, ah scoring for those feature families because a h in from the participants, like when the size is reduced at one point, which of the families get blur?

Speaker 2 01:43

Ah. Or more ah. Hard to ah. Ah identify or which familiesStand out more since these fish families are already in the literature and t hey have been um studies aroundThose.

Speaker 2 01:58

But the gap, what I saw is there's no morein-depth qualitative part where we can see ah with, size reduction of course, the icon get s a bit weird but what.

Speaker 2 02:11

Families stand out ah around that with the proper qualitative score for with the participants so there's some kind of results um th at I'm hoping for along with the yeah technical part of course, but this,

Speaker 2 02:29

is the current current ah working area.

Speaker 1 02:35

So will you be using a certain icon set? Right? And then you um you have your measures, your seven um different like complexity a nd all these color and the measures you showed.

Speaker 1 02:54

But it's always going to be like a combination, I guess, right? Yeah. Um and.

Speaker 1 03:03

You will use one icon set or different I can set, but within an icon set you will present the icons at different scales. And then check, um.

Speaker 1 03:17

What kind of distance humans ah give to pairs of icons? Or will you will you just ask them is it the same or different? What is the g oal there in the in the study?

Speaker 2 03:31

So this is the current area that we have been talking as well. Ah the study quality qualitative study part. Um yes,

Speaker 2 03:41

we had this approach as well that maybe we can have a confidence score along with the all feature families for the participants. A nd of the ah ah like these are theopen-ended questions that ah we have to work on as well but one.

Speaker 2 04:00

Was very this confidence score one ah for, all feature families around the icons and there. Since the users ah have might not have the semantic meaning of all iconsPossible soThis.

Speaker 2 04:16

Is one direction and the other direction we also ah discussed, around since we, have the clustering with the ah our, um, um, distanc es, with the computer vision part and the AI part maybe. We,

Speaker 2 04:32

can also have something similar ah with, the human ah part as well how to. Do they see the distances so ah? Yeah but these are t he current you can say approaches that um I'm currently working on to see how to finalize the quality of study for a good thesis re sult in the end.

Speaker 1 05:05

I think there are different directions where we can go and right now it's a little bit unclear what's what's the most useful one. I gues s the, of course, there is a, um the, we can test the discernibility concerning the size.

Speaker 1 05:28

We can um test discernability concerning the the shape or the the type of icons icons itself in principle, creating and and a new ne w feature that, ah with with the icons that we can add to the meshing features.

Speaker 1 05:50

Um and it's a little bit right now that for me at least, the bigQuestion, what's the most useful direction toGo?

Speaker 1 06:02

Or dora? Am I am I wrong here?

Speaker 2 06:04

Just tell me. No. No, you're fine. Yeah. We are just at the Edge. We are at the Edge of deciding. Like, um well, our our work up till no w was aimed at finding good categories under which we can more or less find good groups of icons.

Speaker 2 06:25

That makes sense to compare. Yeah. So this was like the point until now. The question is, now that we kind of have that, um what kind of study do we want to do and how we want to do it? Ah asking a person given two items,

Speaker 2 06:45

how dissimilar these are difficult.

Speaker 2 06:49

You can say are these the same or not? This is this is easy. Everybody can can give an answer to that. But how dissimilar things ar e?

Speaker 2 07:00

I don't think that we get a good kind of results out of that because everybody has a different scale in mind for that. And you can, it' s difficult to match that. So um it would be like this is the same or not.

Speaker 2 07:16

And if it's like upscale, then of course it's always no. So it's really I don't know whether how we can do it. Another way than really s caling down as much until we have there's no difference left that people can perceive.

Speaker 2 07:34

And the thing that we can get out of this would be different things.

Speaker 2 07:39

Either. We could really get some kind of a weighted measure out of our seven families. Um that best matches what people, how p eople. Oh no. That best can simulate or model how people perceive those items.

Speaker 2 07:58

I can so that we can have an automatic measurement of whether two icons can be discerned, even if they'll be small, would be po ssible. And the other way is that we yeah compare it to the AI evolution, as at some said,

Speaker 2 08:18

and see who which of the AI um embeddings would be closest to what a person would say.

Speaker 1 08:27

A combination of kind if you kind of ah if you can can find stable weights ah with with when when calculating the other the comput er dimensions, that could be also the golden. Yeah。

Speaker 2 08:45

Definitely. So these are the possibilities that we have there. But regarding with the task, that's the question. Um I mean, I don't see another way then with really showing two icons as small as possible and asking whether it's the same or not, or given a very small icon and a set of large icons telling which one is it.

Speaker 3 09:10

Okay. Yeah, yeah, yeah. Um and one issue is like selecting your representatives. If you don't have a very small icon set, then you nee d to select representatives right for that come from the clustering.

Speaker 3 09:27

The clustering uses a distance measure that is kind of given and which is not human calibrated. So the selection itself is an issue of the representatives of the different cluster nodes because the distance measure is not what humans would associate.

Speaker 3 09:47

So so so the the selection already biases this test to ACertain degree, obviously.Um.

Speaker 3 09:56

And then the other issue is also.

Speaker 1 10:01

Ah just downscaling the the pixel size. And maybe you could we could say that this is like the standard way of doing it and this is why we are doing it. But per se, it's not not the only way to downscale just, you know, ah.

Speaker 1 10:23

Putting some filter on it and and telling the downscaling algorithm, scale it down, right? You could emphasize edges or whatnot in the downscaling process. And we know that, for example, color, when I can get very small,

Speaker 1 10:41

um that color is less discernible at some point, especially if it's not like red, green, and blue, like very bright ah colors. So these kin d of things interact. So yeah.

Speaker 1 10:58

Um I like the principal idea of last just showing two icons and and asking is it the same or not? Um about ah.

Speaker 1 11:12

It depends on the preselection of which icons use. If you have these huge icon sets, it's very difficult. And then the problem is also my feeling is that um the aim to find a global waiting.

Speaker 1 11:33

Across the different um distances. You know, I think seven distance families or distance measures or that this is not right. Ah that' s my, my my concern. Um if we have like a set of like faces, right?

Speaker 1 11:51

You have like twenty different faces and ah you would use your ah measure probably. And and and and the the weights for that m easure come from some study of other icons. it's not going to work um because all of them are browned,

Speaker 1 12:12

all of them are yellow. Or the ah the difference. The main differences are likeThe maybe the strokes of the mouth for the nose and for the eyes and the ears whatnot right inThere.

Speaker 1 12:29

Um and.

Speaker 1 12:35

I, my my feeling is that, and and then we have these icons that that are so different, right? Where we have like ah field areas with lot s of details and then you have just some lines, right?

Speaker 1 12:50

And that like a global measure that just assigns constant weights to the different distance measures and then tries to create a gl

obal distance measure from those is not going to work. That's my general feeling.

Speaker 1 13:07

And we are seeing that already because you're doing this to perform the clustering. And if you look at theClusters。

Speaker 1 13:19

In some areas you think, yeah and all those are dense and all those are lines that has worked well in other areas. You see here's a face, and there's a face and some that are really exact same shape, only slight difference in color are not very close together. They're at a certain distance, and so on. So。

Speaker 1 13:40

I'm wondering if the aim for finding a set of global weights to weigh the different distance measures and put them together in here is the measure for your eigenset. That's not the right goal.

Speaker 1 13:58

That's my feeling because it can't work. Because different I can sets up, need different measures. That's my, my, my general feeling. I don't know if I'm right, but.

Speaker 1 14:15

I think I am because currently you ah I have seen that you have like sliders where you can set the weights for the different distance features. And depending on those weights,

Speaker 1 14:30

currently you get a certain map you know of the icon space, and with some weights it works better. Maybe for one I can set with other weights. it works better for another one. But these are also global weights. And somehow。

Speaker 1 14:52

That makes me worry a little bit that.

Speaker 1 14:57

Global distances.

Speaker 1 15:01

That are just computed in some rigid way might not work well in the end. So okay, enough of that. What do you think of that? Do y ou have hope that.

Speaker 1 15:17

This works. I mean the calibration approach is really good, but.

Speaker 1 15:23

What do we learn from the calibration approach?

Speaker 1 15:30

Is maybe maybe weights for these global feature distances, but.

Speaker 1 15:37

Is that it's kind of averaging over all kind of different glyphs and and so on. I'm I'm wondering if it's going to work, so please.

Speaker 1 15:48

Convince me that this is going to work.

Speaker 2 16:00

Yeah for for now, the blotting of the icons, glistering one, ah it's um one forty icons. So we're not plotting them. Ah we just picking them ah in a normalized way. You know with ah stratification, ah randomization, ah and.

Speaker 2 16:28

From our justly gut, gut feeling it it looked good. And because initially when we did not have enough filters, and ah we were just doi ng all random. And ah it was quite it was in very bad formats compared to this one.

Speaker 2 16:50

So with small adjustments it visually it looked good. So maybe we we thought that, yes, it's it's now better than before. So for exa mple, if you see this specific area for the flags, ah it's this one is anomaly,

Speaker 2 17:11

but the flags previously, we never got this kind of um icons together. So um butYeah, that's the only thing that comes in myMind.

Speaker 3 17:32

That's just a small comment. it looks like a very good map of stylistic similarity. But um from a design standpoint, I would say this is very nice to make. Um.

Speaker 3 17:48

Sets of icons that are good together. Um.

Speaker 3 17:55

But I think there are other factors. For example, the semantics, as as professor forge was saying, that's a very human factor that pl ays a big role in discernability. If I look like at the icons that are on the top right,

Speaker 3 18:11

they're very complex and very similar, but still very discernible into what they are. I guess in small sizes not. And then I see like the line icons on the on the bottom left.

Speaker 3 18:26

And I think that a user would have a lot of problems of discerning them because the meanings are not so straightforward, right? S o probably memorizing them and and recognizing that they are different doesn't work as well. Um so maybe considering.

Speaker 3 18:45

Or or kind of contrasting how discernible they are with like these types of measures. it's it's a way to go, like like taking two that ar e close together and trying to see how that compares to this distance in a way。

Speaker 3 19:09

Is something worth exploring.

Speaker 4 19:15

Do you mean?

Speaker 4 19:18

Like seeing when the or maybe I did I did and got it wrong. So discovering。

Speaker 3 19:26

I know, this hidden factor of of discoverability that you can't just measure with computer vision. I mean。

Speaker 4 19:36

What would be interesting if we if we reduce the size, ah letting the people decide so offer, ah which at which they crash. And you can't can no longer see what what it is. And then look at these icons.

Speaker 4 19:59

Um.

Speaker 1 20:00

How they previously have performed concerning the different measures.

Speaker 1 20:07

And then maybe we will find that reduce it, that that items that have a certain profile being quite high in one measure, um that may be one of these measures is good to good predictor of how fast or ah how fast they crash or how they how long you can see the m if they if you're making them small.

Speaker 2 20:37

I think rene made a good point. If you look at this, this is like a stylistic um clustering somehow, you know, that's what you said, rig ht? Like if you look at, especially at the bottom, there are some the stylistic similar and the flags are stylistic similar, but it doesn't mean that the flags are not.

Speaker 2 21:04

Easy to differentiate. They're probably very easy to differentiate, right? So should should should they be far apart or not? Right? So they are semantically, stylistically similar, right? But they are easy to discern, I guess,

Speaker 2 21:21

like the swedish flag and the french flag, I guess those are the two. Um what's the problem to differentiate even if they get smalle r and smaller and smaller and smaller? If you have a blue pixel, a white pixel,

Speaker 2 21:37

and the red pixel, and then yellow and and blue is very easy to, um soThis, and then then we have on the bottom, um we have some that have quite these two blobs,Right?

Speaker 2 21:50

The the light is it slightly greenish? I can't even say, ah have to make it larger. Yeah. Like this area. The the brown on the bottom, on the bottom, on the very bottom, the one that's very on the bottom. Yeah,

Speaker 2 22:03

this one where your crosses and the one on the further right down. Yeah, exactly. So there's this splash two splashes. They're identical in shape, right? And, um, their distance is not that small.

Speaker 3 22:16

Ah and ah。

Speaker 2 22:20

They would be not discernible if they get too smaller because the colors are the the shape is the same, of course, right? The the only difference is the color here. And that gives it such a large distance. And then the red one.

Speaker 2 22:38

I see why it's in the same group. it the, um what is it called? Um like the and can not can but wasn't. What is geese can in english watering? Watering can like,

Speaker 2 22:52

ah you know it it has ah the I don't know what that is called the the owskisa right there where you where the water comesOut.

Speaker 2 23:03

It's in the same direction as with the blobs where the something sticks out from the blob, right? That's probably why it's in the same group, but it's so discernible. it will be forever discernible to these color blobs on the bottom, right. So um.

Speaker 3 23:23

Yeah, I know.

Speaker 2 23:26

I know. it's impossible almost right. And there are some areas where the clustering puts semantically similar things closer togeth er. And then there are some areas where, um.

Speaker 2 23:42

It it it it. Yeah but and and we have to see that these are high dimensional differences, right? And we have like a layout in two DA pr ojects to two d. And that's always but an issue by itself, right? As you know, so it's very hard.

Speaker 2 24:02

To consider these human.

Speaker 2 24:07

To take samples of this, right and and use it in the study and um.

Speaker 2 24:15

To discern them. So how how does this semantic, it's not semantic, but it's somehow stylistic, somewhat layout, right? Ah partiall y at least. Right. Um like the the the sausage. And and then there's the ham and then there's like, ah.

Speaker 2 24:39

So in organ near the the hand, the sausage, and there are similar styles in drawing as in the pointed out that that's very well organiz ed.

Speaker 3 24:52

Um.

Speaker 2 24:55

See. So so here we don't have like a dissimilarity distance.

Speaker 1 25:02

Sometimes we have a semantic similarity, but semantic semantic style similarity, not content similarity.

Speaker 1 25:11

We have style similarity. You know that and and style similarity um is that is diet similarity a good distance measure? Maybe not. Y eah but we we cannot solve that problem, we cannot solve that problem.

Speaker 1 25:28

We might have to we might have to think about it differently somehow, but I don't know. Ah you might have to think of what's the g oal of the study.

Speaker 2 25:45

We were already thinking that we would never return from the cliffs. I mean, these icons were all there for me in the beginning at le ast to find out whether one could have nicely describable groups of icons like these are faces,

Speaker 2 26:04

these are humans, these are whatever. But this ah that's unfortunately not as easy as I would hope to do. So um what we thought and what I think makes sense here is that we get into a smaller glyph sets that have the same design are from the same designer s,

Speaker 2 26:26

have the same the sameLanguage, like visual language also, so that we narrow this down alreadyThere.

Speaker 2 26:35

Because if we match two different styles, as you say, then a, it's discernible. Definitely, because the two different styles, it will be di scernible even if it goes down, if it's not,

Speaker 2 26:48

if there's not a very strong similarity in some other sense there. So that makes sense. Anyway, and it's also very much closer to th e actual um use case, right? In the use case,

Speaker 2 27:01

you have an icon and icon set that somehow connects in a way that is from the same, has the same topic, has the same designer as the same sameLanguage in AWay.

Speaker 2 27:13

And from there perhaps you can get somewhere so that we first never down to a glyph set of twenty. Perhaps that is a similar des ign. Has one topic. And then from there we could. Yeah. Then then we definitely need to set the actual goal. I guess it will not tell u s the goal, but it gets one step closer to that, I hope.

Speaker 1 27:46

Yeah, yeah. Maybe making it more simple. Don't aim for classifying thirteen thousand icons and and come up with a single measur e that across all those icons, and it's human calibrated that is too much to aim for, I think so. Um.

Speaker 2 28:11

Yeah perhaps we can have their categories of icon sets. So very simple. I can set like words, the basic shapes like you know circle s and rectangles and triangles and stuff.

Speaker 2 28:24

And then one that's very complex where a lot of humans are in there, like those that we see here, something like this that we have l ike ah that we define icon sets with different um attributes that we set that we want to find out about where the differences are.

Speaker 2 28:44

And then within those icon sets, we can then um ask for differences and seeHow people behave like withinThose.

Speaker 2 28:52

But yeah yeah what people think to be similar.

Speaker 1 28:57

Yeah and I agree. But what will happen is what I fear. I fear is that, um, if you then have your distance measures and you find some weights such that those distance measures fit as best as possible to the human, um,

Speaker 1 29:17

distance measures, say, um, then you go to different, I can set, and it will be different ways. Um, um, or, ah, you know, um, possibly

。

Speaker 2 29:31

Yeah.

Speaker 1 29:31

And then, how does it generalize then from there? Um ah but we can go this experimental way and and try and see how different th e weights are um that that give a good fit to the human distance measure. Um.

Speaker 1 29:53

I don't see any other maybe a way. Yeah. Public. Yeah.

Speaker 1 30:00

And be alright that most probably we most probably we will have ah different measures for different icon sets. And maybe that's t he stylistic input in principle.

Speaker 1 30:14

Or it or it represents to some degree the stylistic influence and.

Speaker 1 30:21

Which is not bad, but it's hard to compare it then.

Speaker 2 30:27

So the generalization, what you want is somebody, you know somebody that designs an icon set, right? And then what we want is, yeah, here's our measure. You can check if your icons scale well. And then, you know,

Speaker 2 30:44

if the distances and the thing is, even if to if you have a larger icon set and say, and and some icons are confused, which are sema ntically similar, you don't care because it's in the family of friendly faces and you know,

Speaker 2 31:01

of angry faces. And and so if those kind of meld together, fine, right?But ah friendly and angry should not kind of metTogether.

Speaker 2 31:11

Right? So that's what the distance measure needs to capture and that yeah, that's about faces. But if you have like and.

Speaker 2 31:25

For other things. I remember we had, ah patrick, you came up with this set at the beginning like this like black and white, like arrow heart and and all these icons. I don't know what the icon set was from, like from a game or something, right?

Speaker 1 31:46

How we came up with it. I actually don't remember it.

Speaker 2 31:50

Um so it came from a game. I think that wherever that was used. Pretty sure. Yeah, I know. it's like five, six years ago.

Speaker 1 31:58

No, no, no. it was it ah it was the design of mux. Pretty sure. Now. So it was the。

Speaker 2 32:06

Icons that came from some or he selected it from some game. it was he was from he did not make those icons. No, he did not dr aw. He wasn't drawing those icons. He he was like, taking them from ah from a game. Yeah. I remember that. That was like, ah like 。

Speaker 1 32:25

And it was a list was destroyed. And that was yeah, that was exactly. There was a yeah yes, exactly. Yes. A lightning bolt. Yeah, exa ctly. So there was some something that was yeah, yeah, yeah.

Speaker 2 32:42

But he came from again anyway. And and could you please ah look at the the icon sets that you chose that you started from you y ou showed them at the beginning. There was a slide that showed the icon sets. Can we go to that um slide?

Speaker 2 33:03

Yeah, yeah, yeah. We have like some that are not too big, right? And and where it's really important, like the hazard pictograms, it's really nice if they would be differentiated, right? Or the medical devices, the safety signs,

Speaker 2 33:22

um it's really important, right? Um and then, ah the usp medication pictogram is probably also important. Um but ah these ah aras ah as AAC. Pictograms are like, wow, these are really big. Open emoji. Open emoji. Okay.

Speaker 2 33:40

Um these are really, really big. I find even if you onlyFocus on the right side, those seem to beImportant.

Speaker 2 33:51

Ah pictograms, right? Very important pictograms where differentiation, there should be no misunderstanding about which medica l device is this or something like that, right? So um we could look at that and then see as a goal, um which.

Speaker 2 34:13

Ah maybe if you if you try to find a way to a distance measure and and match it to the human perception, um that's good. We can d o that and have a study with all those on the right, say, and the selected candidates from that,

Speaker 2 34:35

and um then calibrate our measure. But my concern, and maybe I'm wrong. That would be good is that um ah the weights would be similar across those. Not too different. Um and we can then calibrate to each of the datasets,

Speaker 2 34:57

then compute like an.

Speaker 1 35:00

Average measure and then check how bad is the average measure measure to the optimized measure? And that would be a really great result of the thesis. it can do that, but maybe that's too much work.

Speaker 1 35:14

I don't know if that can be done, but ah I think that will be a result right? Either the average measure across the different ah data s ets is ah is effective compared to the individual measures, then that would be good result,

Speaker 1 35:30

perfect result. If it's not defective, we know that.

Speaker 1 35:36

It's not. So we already know it's not so easy. But but we can say no, we have to find as a result of thesis, yes, we can have individual measures that are effective if you calibrate with humans,

Speaker 1 35:51

but generalizing that it's it's difficult, right? That would be also good. it's not your fault, right? If if that would be the result, right? But we at least arrive at can you generalize or not? Or even if I can say if to ah you,

Speaker 1 36:09

ah all we have or then the way forward would be not in your thesis, but the futureWork would beOkay.

Speaker 1 36:17

That different types, stylistic types, as in the quality of icon sets. And if you have measures in those stylistic types, um okay. Which you you design a new icon set look,

Speaker 1 36:31

um ah to which one is it most similar of those example I can set. And then take that measure, try that measure and see it could be a good fit, right? Or is it between those two?

Speaker 1 36:45

Then take an average of those two measures that be good for those two icon sets, right? And I think there are not too many different, ah so that many differentStyles, but but maybe there are more like like fill field styles,

Speaker 1 37:02

and then there are more line styles and black and white styles and and and and soOn.

Speaker 1 37:09

Maybe that that helps a little bit to organize that space. Having different measures. And and maybe there could be even an algorithm that kind of first checks um your new icon set to the already calibrated icon sets to which is it most similar in some sense?

Speaker 1 37:29

Right? And then use that measure instead of like have one global measure for all. I can set that that that we can imagine. And I thi

nk that's a good way forward from here. But and what you mean is all at once. Yes.

Speaker 2 37:46

But what you mean is in the tests that the people should, ah.

Speaker 2 37:54

State a similarity between the different icons.

Speaker 1 37:59

Or we we we can say we, but we can if if it's mostly about ah ah mostly about um the different, we can we can focus on one thing, scalability, right? Then ah we we can just do that and and then just set the yes, no, because if people really have to rate the similari ty between them。

Speaker 1 38:31

Um yeah dora is cooking something and it's burned. Okay then. But now, yeah, but this is the working. Yeah。

Speaker 2 38:43

Yeah. So the scalability。

Speaker 1 38:44

You have to have one goal. Yes. Let's let's just first look at scalability. I don't see any. Let's see. Let's say I can. Designers would um aim for having a set that it has reasonable differences at a normal scale, at a good scale.

Speaker 1 39:04

And then, ah the interesting aspect is how it does it scale? And then let's let's look within one egg I can family. And then calibrate i t to a human scaling the perception,

Speaker 1 39:20

and then see if we can come up with a weight for our distance measures, different distance measures thatEmulate or approximat e or model um the human distance measure within the different icon sets weInvestigate.

Speaker 1 39:39

But I don't know if would it be possible to run the study on um a sub sample of those icon sets on the right here, say.

Speaker 2 39:52

I mean。

Speaker 1 39:52

That's it. Or if that's just too much。

Speaker 2 39:58

Where we have to think about how.

Speaker 1 40:00

We want to present the scaling. So ah what what steps do we need? So how many steps people should assess or should they use a slider and say, that is the point where I can no longer identify it?

Speaker 1 40:16

Ah what's the what's the the best result? And that factors into how many slides we have to show to the people on prolific or in, in or not, not on prolific and with the people. So, and then it,

Speaker 1 40:31

that kind of factors in how many icons we actually can show. Um and so here we have eighty eighty three.

Speaker 1 40:40

So that's one hundred and sixty. So that would be three hundred, three hundred and sixty icons if we focus on the right side.

Speaker 1 40:51

And then we want to crash all of them with people. That's you know that's yeah that's like subway.

Speaker 2 40:59

Of course. Of course we have to select subsets right for the nine.

Speaker 1 41:03

So do that. Yeah. So so dora, do you remember how many ah.

Speaker 1 41:10

In how many slides we showed in the in the rose charts? Hundred and twenty. What was that? The reasonable. That was a reasona ble size. Yeah people to look at at, to assess and to put some value in. Yeah.

Speaker 1 41:28

Um I think if we go in that direction saying we want to show one hundred and twenty icons and then we, we, we let I don't know wi th that we, so the。

Speaker 1 41:42

The, the whether they should do a slider on their own or the weather, we show them different sizes that we have to think about. I d on't know what is the best, what's the best approach here.

Speaker 2 41:58

Maybe you have to test, right? Both slider takes time because people go back and forth, back and forth and so on. But showing m any different conditions like to narrow it down to a reasonable scale. That's also a long process, right? And you probably have to h ave to balance it for order.

Speaker 1 42:21

So I mean。

Speaker 2 42:23

If we can be random。

Speaker 1 42:24

If we have one icon and then decreasing sizes of that icon, so with ah with, ah, um and then they have to pick which one they can't see. They can't identify anymore.

Speaker 3 42:41

Um the problem with that that I see is that they know what the large icon is. And so it might be that they can discern it further as t

hey actually can discern it because they know what it should be. And the yeah or。

Speaker 1 43:01

We do it the other way around that that that they have a slide or they cannot ah they cannot see what it is. And then at some point they see it and then they have to lose leave the slider in that position saying, okay, now I see it.

Speaker 3 43:19

Or it's like iterative that they have these icons as long as they can't discern it, they press like Plus until they can and then they have to do next or something like this, something like this.

Speaker 1 43:32

But it's like, I mean, that's in principle the same thing.

Speaker 3 43:35

It's in principle the same.

Speaker 1 43:37

Yeah so if yeah but we can have the sliders having discrete positions thing. Okay now。

Speaker 2 43:43

But the thing is that.

Speaker 2 43:49

If you put them next to each other。

Speaker 1 43:51

It's no you don't put um you don't put them to each other.

Speaker 2 43:57

Yeah. But where do you put them in the?

Speaker 1 43:59

No, if you if you have the slider, you start maybe just with a black dot danger. But you.

Speaker 2 44:06

Yeah。

Speaker 1 44:07

You put the right slider to the right, then it becomes bigger.

Speaker 1 44:12

Then it becomes bigger. And then at some point they they they can identify it and then that's the position. But it doesn't like。

Speaker 2 44:19

Ah okay, you you you don't want to have discernability. You want one identity, viability. That's something different, right? it's true.

Speaker 3 44:31

That's what I was looking at. Yeah. Yeah.

Speaker 1 44:33

But but for me, I thought that is the first step actually, but we can discuss about it. Of course. Um I think that that's what what wha t you meant on but done. So. Yeah。

Speaker 2 44:49

Yeah, yeah. Ability. Yeah. I was thinking really, um ah key just differences between two. I was at first thinking, you know.

Speaker 1 45:00

Like you scale both of them up and down, right? Or you start from towards two dots, scale them up and then you look at them and then, oh, now I can see they're different, right? They you don't know what they are, but you're right,

Speaker 1 45:16

the semantics of an icon is important. it's not only that they different, but you need to recognize them as well. So yeah, that recog

nition and difference is both important, right?

Speaker 1 45:29

And then we I had the example before between the angry faces and the friendly faces. Ah yeah,That's the that's a different aspect, Right?

Speaker 1 45:39

Different aspects are different. Similar. Yeah. Semantic difference, basically。

Speaker 2 45:44

That's the difference between those two.

Speaker 2 45:49

There are two laws for that, but I cannot. The one is the there are two laws for that is is ah seeing that there is a difference and kn owing what it is. So there are two different laws for that, but I have it somewhere. Good.

Speaker 3 46:10

Okay.

Speaker 1 46:13

What do we aim for? Is then the question, right? Um ah you're you're right with, I can say it's identifiability, but it's very important.

Speaker 1 46:23

But also differentiability is important. Both is important, both are important, but maybe those are the most important access, righ t? So that would be two access.

Speaker 3 46:36

Um.

Speaker 3 46:42

I mean for um.

Speaker 4 46:48

Um ah then we would need two tests. So would it suffice to have once? I mean, for patrick's test, we actually don't. And only need t he scaling up,

Speaker 4 46:59

but also like a list of all icons in the set so that people can actually select which one it is. They think it is. And the question is whet her people then won't go to be sure and go even higher than they needed to be.

Speaker 5 47:23

You could do it as a tool to alternative force choice where you show them only one scale and they can. They decide, I know what it is. I don't. And at some point, if you repeat that in random order with different icons and different sizes, you get like a point or a sc ale where they mostly know what it is and mostly don't.

Speaker 1 47:50

But remember, um the people do they do, they need to be able to remember those icons then by name or ah how how to do that, ri ght? Um how would they say, yes, it's this icon. If you have those nine hazard pictograms,

Speaker 1 48:10

they can learn those maybe right at the beginning and then say, yeah, it's this one or this one, but if it's more impossible, it's almos t impossible. So you have to ah show them the existing icons,

Speaker 1 48:28

ah so they can select from those, otherwise they have to remember those from memory.

Speaker 1 48:40

It's difficult, really difficult.

Speaker 4 48:47

Another possibility that came to my mind, which is not about scaling actually, but one could ask people for similarity in a way that we have like one target icon in the center and then one another one to the left and another one to the right,

Speaker 4 49:01

and they select which one is more similar to the center, one. That would be a possibility without scaling. Then we don't have the s caling part, but we have some kind of a similarity measure. Yeah。

Speaker 3 49:12

Yeah.

Speaker 1 49:21

That would, for example, if you have those faces, angry and friendly faces, it would create those clusters, very obviously. Um defini tely。

Speaker 3 49:29

Yeah。

Speaker 4 49:33

But then we don't have this the yeah the discernability in there.

Speaker 3 49:37

But yeah。

Speaker 1 49:39

I mean。

Speaker 2 49:41

We we we could make two tests, two pilots, and one was scalped, one was identifiability and one was discernability.

Speaker 2 49:55

I mean it in principle, we could do two small talk pilots.

Speaker 1 50:00

And see what is more worth while to pursue.

Speaker 2 50:08

I wouldn't. I would have even problems how to set up the test, right? Identifiability. Ah would be you have the slider and your scale and then you have all your icons around. And then at some point you say you, you,

Speaker 2 50:27

you look at them, right? And then you scale it up. And then if somebody, ah it's this one, right? So, um but then if you're wrong, wha t does that mean? Right? You missed. Okay. Ah so you didn't select the right one. Good, good,

Speaker 2 50:47

okay?

Speaker 2 50:51

You could say scale up until you feel safe to select which one it is, so that would scale up more.

Speaker 2 51:01

But is that is that a good? Is that a good measure for discernability? I'm wondering, right? Or identify identify the ability, right? Ah i t's this one, okay? it's now another one. You do the same with another one.

Speaker 2 51:25

Um and you know, and then, yeah. Ah but but what does it mean? How do we create a measure from that? That? That's my questio n right now. We have then, you know,

Speaker 2 51:39

ah which icons people matched well and which icons people don't match well or only match at a larger scale or whatever, that wo uld be the result of the study, right? So how what do we do with that?

Speaker 2 51:56

How do we create something of that?

Speaker 3 51:59

I mean from the from the icons that don't need to upscale a lot to be identified. I guess those are very pregnant. We don't do. We k now that these have good quality for discernability and those where you have to scale very often,

Speaker 3 52:15

although so we need a large scale from them. They are too either too complicated or too similar to some other icon. I guess that's all that you can say more or less than that. Yeah.

Speaker 2 52:28

Yeah. Yeah. And but but how to create a measure from that? The goal is kind of this measure in the end.

Speaker 3 52:34

Right? Yeah. True. Yeah.

Speaker 2 52:37

How to do that. Yeah.

Speaker 1 52:39

I mean, isn't the measure if so, if we if we see for each icon?

Speaker 1 52:47

Yeah. Okay for I can I have an idea for the identifiability. So if we have this identifiability score, then we could look for a linear combination of the other seven.

Speaker 1 53:04

Ah that would be a principle that would deliver the weights. Um but II don't know how we do it with discriminability. I mean, we have to differ. We have the difference between two eigens concerning the other seven scales as well.

Speaker 2 53:52

Yeah, I don't think we solved this now. And let's think about it. And at least we have two two things. The one is ah so maybe not two things, but it's like with respect to scalable,

Speaker 2 54:05

we are interested in maybe we're interested in scalability. Maybe maybe not. That's if we have a fixed size, we could then use another fixed size or whatever. But if you have a fixed size, and the question is,

Speaker 2 54:21

is this a good icon set or not?

Speaker 2 54:27

And and good means that I can buy it identifiable and discernible or discriminate discriminate them. The the the these are the two things okay?

Speaker 2 54:43

So far we got and now we have to.

Speaker 2 54:50

Develop a study task from that, or or how if you do a study looking at these kinds of.

Speaker 1 55:00

What can what can we do with the results? What do the results of the study tell us? Um and how can we use that for future icon se ts? For the design of future icon sets of ah better maybe even for the evaluation of future icon sets.

Speaker 1 55:18

This is how far we got. And I don't think we get any further today and we have to kind of think about that. That that's my my feelin g right now. Yeah.

Speaker 1 55:34

And we decided, I think we will focus on small icon sets or small samples of large icon sets to make it simple. Um but we have to revisit really the question.

Speaker 1 55:50

And what we can do with the results. And the the question maybe is discernability and identifiability. That. But that should be also i n the literature already like captured.

Speaker 1 56:04

Then how do we turn this into a study and what do we do with the result of the study? That's the main challenge I think because y

ou want to use the results for something, because otherwise they're not very useful.

Speaker 1 56:21

If you just evaluate an existing icon set, and then we say, yeah, and some of the icons are well discernible, others areWell identifiable. That's not. What can we do withThat?

Speaker 1 56:35

Nothing but ah, ah so um.

Speaker 1 56:42

Even if it's very basic, what we get out of this in the end.

Speaker 1 56:48

It should have some at least a guideline for designing items. That's the minimum. I would say if we investigate some icon sets and and and get something out of, we have a recommendation. And the ideal thing would be,

Speaker 1 57:05

we have a measure somehow that right? That would be. But that seems to me now that we talked about it for so long, it seems to be a very hard thing. I yeah. And it's not at all.

Speaker 1 57:19

Ah your fault your fault ah that you implemented already a lot. And that brought us to the core of the problemSomehow.

Speaker 1 57:29

Right? But what you did so far, and that's very valuable. So don't worry about it. it's not your fault at all. If you don't come to a good, good, easy result in it, easy. it will not be an easy result.

Speaker 1 57:43

it will be ambiguous in any way, but we have to ah come to a conclusion. What is worthwhile investigating and how. Um and I don't think we get to that right now, but maybe we will discuss in our round at some point about it.

Speaker 1 57:59

And you can discuss with dora and patrick and maybe you're in.

Speaker 1 58:04

And she also can join in um and and think about it and and then come up with something where the results of the study means so mething.

Speaker 1 58:18

I'm not sure what right now exactly. But yeah, that's the goal. And don't be too too ambitious be. I think it's better to have a small r esult. Like here's a guideline, how design and I can set.

Speaker 1 58:34

And maybe there are already guidelines, but one that's motivated by the study. And maybe also what you did already by the cluste ring and all these things that comes out of both. Maybe.

Speaker 1 58:50

And that will be a great result for your thesis. But don't be over ambitious. Like use three thousand icons and try to find like AMea sure that's calibrated for all of thoseSets.

Speaker 1 59:06

I don't think it's going to work. So. Okay. Um maybe these are my final words for today. Okay. Thanks a lot. Yeah, thank you. it's ver y interesting. it's very challenging, right? it's it's like how how to get to result.

Speaker 1 59:25

That is meaningful in the end. That's the greatest, largest, biggest challenge right now, I think. And I hope um we find something e ven we we don't come to a conclusion. I think we we kind of,

Speaker 1 59:42

um but we kind of broke the problem apart. So we we now have a at least know.

Speaker 1 59:52

Why it's not so easy, tractable, right? And that that's very valuable and if that's a result of these is great.

Speaker 1 1:00:00

I'm happy with that too.

Speaker 2 1:00:03

Good then.

Speaker 1 1:00:04

Um I believe, and you can stay if you want. So hey, I have to work on something else. Um we were in our group talk at some point.

Speaker 1 1:00:14

Once you have thought about it a little more and then maybe we have another meeting in one or two weeks to ah nail it. I would sa y we never made one hundred percent. But anyway, let's try that.

Speaker 1 1:00:30

Thank you a lot for your presentation today. And I think it was a very a good and inspiring discussion. So let's see. Yeah. Bye bye ti me. Bye.

Speaker 3 1:00:47

Okay, good. Um so what would be the next steps? I guess what we should look into are the icon sets, those on the right side that we narrow down each of those to how many should we do? Ten or should we go up to twenty?

Speaker 4 1:01:07

The GHHGHS has it once has only nine.

Speaker 3 1:01:13

So yeah, okay. Yeah. So we don't have to narrow.

Speaker 2 1:01:16

So if we if the I think the question is should we mix them really up or not? Also using two, two.

Speaker 2 1:01:29

So let me. Can you show them? This is again.

Speaker 3 1:01:33

Start with the icons. No, I don't think that we should mix the nerves. We should take some of each. The question is how many betw een ten and twenty something um that yeah the question is some that distinguish very well and some that don't.

Speaker 3 1:01:54

And um perhaps even perhaps, we can also say just like those that are closest to each other, that are most similar visually and of t hose from each of these. And we in our study, we always have matches from within one set. I don't would would like to to mix the m up between between exactly.

Speaker 2 1:02:21

So, so that, and and okay, so we choose twenty of each, let's say we ah.

Speaker 2 1:02:32

Let's say we do.

Speaker 2 1:02:37

So how many do we have? Six. So we don't use the the hazard pictograms or do we use them? I don't know. Um and then we have twenty of each of the other ones. That's kind of hundred hundred slides to show. And then the question is, do we do scalability or do we do comparison?

Speaker 3 1:03:07

I guess for first I would say identification, identibility, identifiability, we called it, I think, identifier.

Speaker 2 1:03:14

Okay. Yeah, okay.

Speaker 3 1:03:16

Then let's let's start with that. And then it's actually a hundred slides. Because if we would have comparison, then we would have t o like have more of that, right? Because we need combinations that not each with even each with each actually within those twent y and。

Speaker 4 1:03:36

Those twenty one.

Speaker 2 1:03:37

So I would totally random.

Speaker 4 1:03:41

For each participant or do we fix?

Speaker 2 1:03:43

Maybe ah may i. What what I would say is maybe we could, um if you can show us the links or if you can give us the links again ah of of the of the icon sets, everyone can ah can look it up and pick out the twenty likes. And then we find a we find a common groun d。

Speaker 3 1:04:07

But that that might be a good idea. Yeah but yeah we do our homework for next week. Each of us yeah selects the twenty that the y find most sensible.

Speaker 2 1:04:18

Yeah um ah and the question yeah. And especially the question is what what what guides to that? So is it. Should we focus on how so should we choose similar ones?

Speaker 2 1:04:30

Should we choose what we think are representable icons of that icon set? Should we choose ones that are already seen quite diff erent to us um in in? And then of course if it is a different in terms of semantics,

Speaker 2 1:04:45

it's always a different in terms of.

Speaker 2 1:04:49

Of the icon itself. There's so so many questions。

Speaker 3 1:04:54

So many players and so many questions. Definitely. I mean, these are our factors and be in your thesis, all of.

Speaker 1 1:05:00

These factors should be mentioned at least I think the question is that right? If it's a representative, if you say it's a representative sample, that means more or less that we have more the more different ones, right?

Speaker 1 1:05:16

But since we want to have discernability, or at some point at least discernability also for identifiability is it's it's important to know where the differences are. So.

Speaker 1 1:05:31

It might be interesting to really have those that are closest visually.

Speaker 2 1:05:34

Or what we also could do is we have a look, we we could show ah the small sample sets in ah in in in the web APP, right? Yes.

Speaker 3 1:05:47

I was in principle also thinking the and all we can have A.

Speaker 2 1:05:52

So maybe you could view, maybe you could have, ah um that would be cool if you could make for every of the fives, you could sho w them and maybe make screenshots of where they are in the two dimensional space.

Speaker 2 1:06:09

And then we already can see. And then we can do a kind of a educated sampling of that. Everyone maybe ah makes their choices and then we we on the next meeting we agree. But what we want compare.

Speaker 1 1:06:25

Yeah, that would be a possibility. Yeah.

Speaker 3 1:06:28

What I will do, then I will, ah, ah remove these ones or should I just comment out from the quarter? Should I just remove them? The bigger ones?

Speaker 1 1:06:39

Um yeah I think for now you can leave it in because in your in your view, you can filter by source, right? Yes. Yeah. Then we simply, a h you filter so that you see only each of these separately without any and other, and make a screenshot of each. So that we have li ke the view of only those for each set.

Speaker 2 1:07:05

Yes。

Speaker 1 1:07:06

That's I think that would be that's a good idea. And then we can you can have a look at those like yeah, only one. Oh no, that's that's just one one fifty.

Speaker 3 1:07:22

But I think this is broken.

Speaker 1 1:07:26

Yeah. Okay, I see. Okay. Yeah。

Speaker 2 1:07:29

It doesn't matter. Have a look at it and and where we have to see the embedding of of each of them individually of each icon set. A nd then we we we we we simply and either we can discuss it in our group or everyone draws the twenty he likes. And then we we a gree on the final twenty. Yeah.

Speaker 1 1:07:53

Yeah.

Speaker 2 1:07:54

And then we prepare the, the, the but and then we prepare the, the, the upscaling and downscaling.

Speaker 1 1:08:02

Um we can have a first pilot. I think that's that.

Speaker 1 1:08:11

Yeah, yeah, yeah. I think that would be the next step for for the first study. And then we will see how we do this. The similarity to so rt of similarity.

Speaker 2 1:08:23

And I found the two, the two or no, no, I've closed it again. Ah.

Speaker 2 1:08:31

So it's a very professional, that's where where is it?

Speaker 2 1:08:43

The professional law that's notifying that there is a difference and.

Speaker 2 1:08:53

And then get, then there is so so stone. But so stone is usually more. But that that means making a choice. So which one is which? Um but I think that's not precisely identifiability because then you're just making a choice. Which one is different to the other or or or which one is different.

Speaker 2 1:09:22

But it's not that, but that's not identifiability yet, I guess. But but that I'm not sure about it. Maybe I have to to to to read it.

Speaker 2 1:09:34

Again.

Speaker 1 1:09:38

Yeah because of this。

Speaker 2 1:09:39

This principle is principle in principle identifiability meaning recognizing the semantics or recognizing or is identifiability meaning I so I see now clearly the shape and what the shape shape is, or is it or is it?

Speaker 1 1:10:01

Philosophy colors。

Speaker 2 1:10:02

Or is it is it identifying the transcendent meaning of the shape?

Speaker 2 1:10:12

Which would be kind of semantics. Ah there are so many words. Yeah, so many.

Speaker 1 1:10:25

Yeah, yeah, yeah. Okay.

Speaker 2 1:10:29

I have to I also have to have to go then yeah to the university. So so next next you say, ah it is on you. You are sending out the the s creenshots. Yeah and then everyone will choose. And then we find a common ground. And I don't know. We we also have to think a bout coming ground.

Speaker 1 1:10:51

You said that at the usual slot where we meet eptizam, you have another meeting now.

Speaker 1 1:10:58

So when can we meet? Okay。

Speaker 2 1:11:00

Let me check.

Speaker 2 1:11:04

Ah that that was it was tuesdays at eleven.

Speaker 1 1:11:09

Yeah.

Speaker 2 1:11:18

That's tomorrow. Yeah。

Speaker 1 1:11:21

Yeah.

Speaker 2 1:11:21

I could go out to。

Speaker 1 1:11:23

Ah wouldn't do tomorrow, but like next week.

Speaker 2 1:11:26

Yeah. Okay. Okay. Then and next week, then then let's keep it. I can I I'm now the the lab classes are kind of set up and I'm and II co uld. And usually next week I have two people in the webcast so I don't need to be there. So next week should work.

Speaker 1 1:11:47

Okay. Then let's say next week at eleven, tuesday, and we will compare our results.

Speaker 2 1:11:55

Okay, then see you. Yeah。

Speaker 1 1:11:58

Thank you for your work.